MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MICROPROCESSOR, DIGITAL, KEY CODE HRH

This amendment forms a part of Military Specification MIL-M-28787/237A(EC), dated 17 August 1983, and is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGES 4 AND 5

TABLE I: Delete and substitute new table I as printed on pages 3 and 4 of this amendment.

PAGE 25

FIGURE 10, ITEM P01-1 block: Delete "4.5" and substitute "5.5".

PAGE 32

TABLE XII, Test condition E2, Max column: Delete "100" and substitute "120".

PAGE 33

FIGURE 12, ITEM P01-1 block: Delete "5.5" and substitute "4.5".

PAGE 35

TABLE XIV, Test conditions G1 through G8, Min column: Delete "-0.01" and substitute "-2.00".

TABLE XIV, Test conditions G1 through G8, Max column: Delete "-2.00" and substitute "0.01".

PAGE 37

FIGURE 13, ITEM P01-1 block: Delete "5.5" and substitute "4.5".

FIGURE 13, NOTE 1: Delete "H1-H14; 0.00 ±0.01 V." and substitute "H1-H6; 0.00 ±0.01 V. H7-H14; 0.8 V ±1%."

PAGES 41, 42, AND 43

TABLE XVII, Test conditions J1 through J24, Min column: Delete "0" (three places) and substitute "-100.0" (three places).

TABLE XVII, Test conditions J1 through J24, Max column: Delete "-100" (three places) and substitute "0.1" (three places).

TABLE XVII, Test conditions J10 through J18, Unit column: Delete "ΩA" and substitute "μA".

PAGE 47

FIGURE 15: Delete and substitute new figure 15 as printed on page 5 of this amendment.

PAGE 86

TABLE XXXV: Delete and substitute new page 86 of table XXXV as printed on page 6 of this amendment.

FSC 5963

Source: https://assist.dla.mil -- Downloaded: 202640-20T16:51Z
Check the source to verify that this is the current version before use.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N662)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:51Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B5	Power supply stress check	IV	S	← Must pass all remaining tests →										
C1-C30	Output voltage, low level	V	X	0	450	0	450	0	450	0	450	0	450	mV
D1-D29	Output voltage, high level	VI	X	3.7	4.5	3.7	4.5	3.7	4.5	3.7	4.5	3.7	4.5	V
E1	Power supply current (V_{BB})	XII	X	0	1	0	1	0	1	0	1	0	1	mA
E2	Power supply current (V_{CC})	XII	X	10	120	10	120	10	120	10	120	10	120	mA
E3	Power supply current (V_{DD})	XII	X	10	80	10	80	10	80	10	80	10	80	mA
F1-F6	Input current, high level	XIII	X	-10	10	-10	10	-10	10	-10	10	-10	10	µA
F7-F14	Input current high level	XIII	X	-100	100	-100	100	-100	100	-100	100	-100	100	µA
G1-G8	Input current, data bus, 3.7 V level	XIV	X	-2.00	0.01	-2.00	0.01	-2.00	0.01	-2.00	0.01	-2.00	0.01	mA
H1-H6	Input current, low level	XV	X	-10	10	-10	10	-10	10	-10	10	-10	10	µA
H7-H14	Input current, low level	XV	X	0	-100	0	-100	0	-100	0	-100	0	-100	µA
J1-J24	Three-state output leakage current, low level	XVII	X	-100.0	0.1	-100.0	0.1	-100.0	0.1	-100.0	0.1	-100.0	0.1	µA
K1-K24	Three-state output leakage current, high level	XVIII	X	0	10	0	10	0	10	0	10	0	10	µA
L1-L4	Dynamic functional	XX	X	← See table XX →										
M1-M4	Dynamic functional and propagation delay time	XXV	Q	← See table XXV →										
N1-N30	Output transition time (t_{TLH})	XXXI	Q	3	35	3	35	3	35	3	35	3	35	ns
P1-P30	Output transition time (t_{THL})	XXXII	Q	2	25	2	25	2	25	2	25	2	25	ns
Q1-Q3	Power supply transient amplitude	XXXIII	Q	0	400	0	400	0	400	0	400	0	400	mV

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:51Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
R1-R3	Power supply filter capacitance	XXXIV	S	4.06	7.54	4.06	7.54	4.06	7.54	4.06	7.54	4.06	7.54	µF
S1,S2	Input capacitance	XXXV	Q	2	35	2	35	2	35	2	35	2	35	pF
S3-S14	Input capacitance	XXXV	Q	2	20	2	20	2	20	2	20	2	20	pF
T1-T22	Output capacitance	XXXV	Q	2	20	2	20	2	20	2	20	2	20	pF

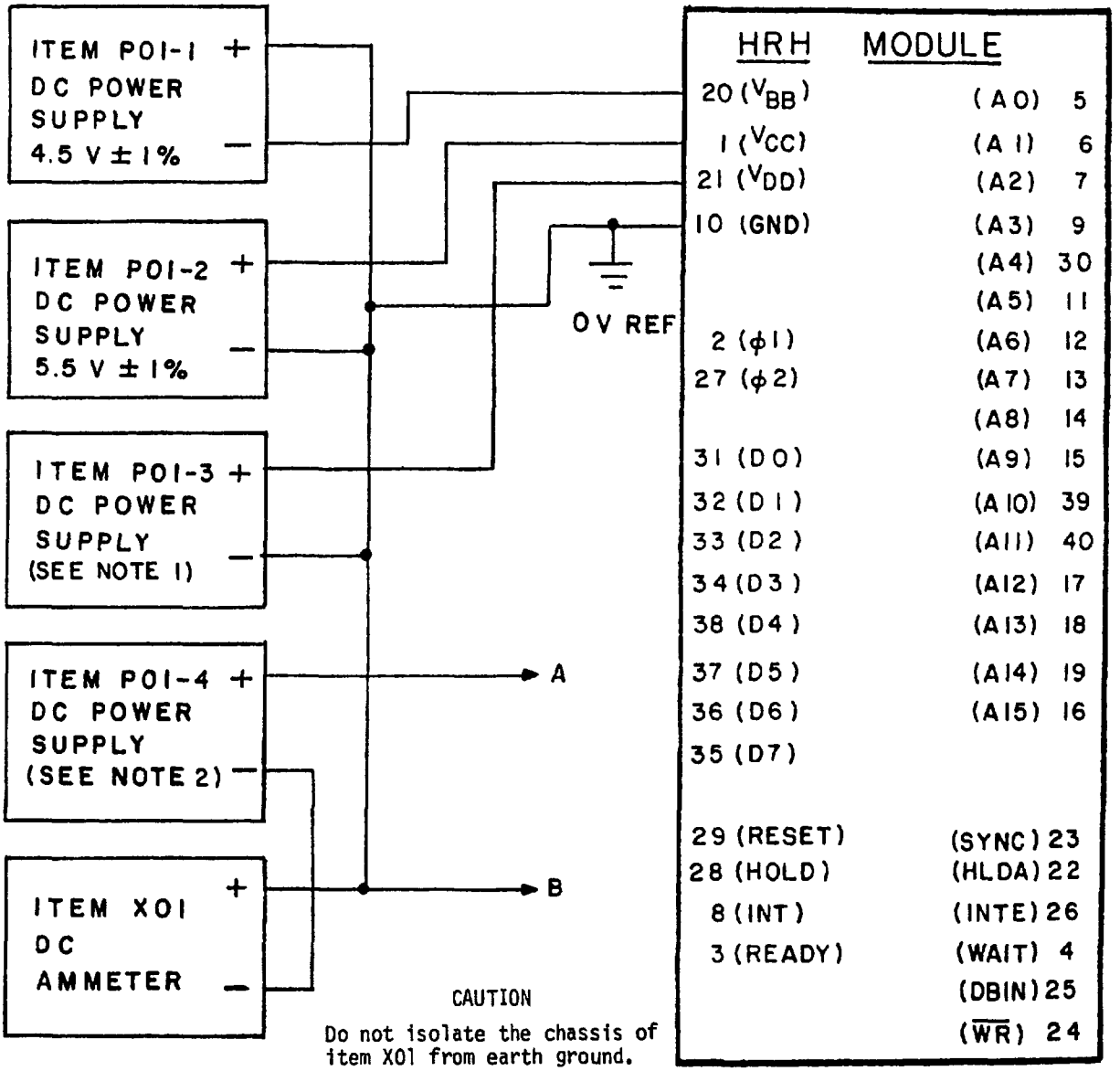
1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:51Z
Check the source to verify that this is the current version before use.



NOTES:
1. Item PO1-3 setting for test conditions:
   J1-J24; 13.2 V ±1%.
   K1-K24; 10.8 V ±1%.
2. Item PO1-4 setting for test conditions:
   J1-J24; 0.45 V ±1%.
   K1-K24; 5.5 V ±1%.
3. Apply power in the following order: V_{BB}, V_{CC}, and V_{DD}.
4. Apply the preconditioning pattern in table XIX using the timing conditions of figure 14.
5. Continue the waveforms and connections after the last vector is reached. Make the indicated connections at points A and B and make the measurements.
6. Since the measurements do not change the state of the microprocessor, measurements can continue on succeeding pins without reapplying the preconditioning pattern as long as the connections and the clock waveforms are continued.

FIGURE 15. Three-state output leakage current (low-level and high-level) test circuit for tables XVII and XVIII.

Source: https://assist.dla.mil -- Downloaded: 202610-20T16:51Z
Check the source to verify that this is the current version before use.

TABLE XXXV. Input and output capacitance test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
S1	O1	Input capacitance	2	Open	2	35	pF	31
S2	O2	"	27	"	"	35	"	"
S3	DO	"	31	"	"	20	"	"
S4	D1	"	32	"	"	"	"	"
S5	D2	"	33	"	"	"	"	"
S6	D3	"	34	"	"	"	"	"
S7	D4	"	38	"	"	"	"	"
S8	D5	"	37	"	"	"	"	"
S9	D6	"	36	"	"	"	"	"
S10	D7	"	35	"	"	"	"	"
S11	RESET	"	29	"	"	"	"	"
S12	HOLD	"	28	"	"	"	"	"
S13	INT	"	8	"	"	"	"	"
S14	READY	"	3	"	"	"	"	"
T1	A0	Output capacitance	5	"	"	"	"	"
T2	A1	"	6	"	"	"	"	"
T3	A2	"	7	"	"	"	"	"
T4	A3	"	9	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:51Z
Check the source to verify that this is the current version before use.